Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	31 January 2026
Team ID	anshrathore0202-cyber
Project Name	F1 Racing Intelligence Analysis
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the technologies, platforms, databases, BI tools, and integrations that you will use to build your Business Intelligence solution. A well-chosen technology stack helps ensure that the solution is scalable, maintainable, secure, and suitable for data analysis and reporting requirements.

S.No	Component	Description	Technology	Justification / Purpose
1.	**Programming / Query Language**	Language used to query, filter, transform, aggregate, and analyze data.	**Python, SQL, HTML, CSS**	Python (Pandas, NumPy) enables high-performance batch cleaning of telemetry data; SQL allows relational querying; HTML/CSS formats visual reports.
2.	**Database / Data Warehouse**	Warehouse/Storage used to store and manage structured business data for analysis and reporting.	**CSV Datastore, SQLite / MySQL**	Structured CSV local datastore in VS Code path (`C:\Users\HP\Desktop\F1-Analysis`) allows fast, lightweight, and offline multi-season relational analysis.
3.	**Data Ingestion**	Used to load data from source files or external systems into the analytical environment.	**Tableau Data Connections, Python Pandas ETL**	Direct live file connectors in Tableau coupled with automated Python ingestion scripts load raw Ergast/F1 racing records seamlessly into memory.
4.	**Data Transformation and Business Logic**	Tools/Logic used to clean, join, filter, aggregate data for analysis and calculate KPIs, metrics, business rules, and analytical measures.	**Tableau Desktop, Python ETL Scripts**	Permits calculated fields (qualifying vs finish position delta, stint pace, pit stop efficiency) and cleans null values before visualization rendering.
5.	**Data Visualization / Analytics**	Used to create interactive dashboards, charts, reports, KPIs, and visual analysis.	**Tableau Desktop, Python (Matplotlib/Seaborn)**	Delivers interactive visual dashboards, driver head-to-head lap plots, heatmaps, and multi-season constructor points progression charts.

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6.	**Front-End / User Interface**	Interface through which users interact with dashboards, filters, parameters, and visual insights.	**Tableau Public, VS Code Environment**	Provides an intuitive, web-accessible dashboard interface for fans and analysts to interactively filter F1 statistics by season, team, and circuit.